KENNESAW STATE UNIVERSITY
COLLEGE OF COMPUTING AND SOFTWARE ENGINEERING
*School of Data Science and Analytics*

# Signal Detection Theory in Black Holes: Extracting Information from the Edge of Spacetime

AJ Savarese – Spring 2026

Dr. Bell

## INTRODUCTION

Black holes can exhibit X-ray emission that is linked to optical outbursts. Using the BlackCAT catalog, I wanted to investigate whether brighter outbursts ("signal") are associated with stronger X-ray flux and higher detectability, while accounting for distance ("noise"). My hypothesis follows the Signal Detection Theory (SDT) framework.

**Stronger outbursts = Stronger detectability**
**Greater distance = Lower detectability**

## METHODS

**Data:**

- BlackCAT black hole catalog
  - 73 observations

**Variables of importance:**

- Magnitude (quiescent & outburst)
- X-ray flux
- Distance

**Data preprocessing:**

- MagDiff = MagOut − MagQui
- log-transformed flux & distance to reduce skewness & scale small values
- logSNR = logFx − 2*logd (used this as proxy for detectability)

**Analysis:**

- Descriptive statistics
- Scatterplots to visualize relationships (Figures 1 & 2)
- Pearson r correlation analyses (Figure 3)
- Simple & multiple linear regression:
  - logSNR ~ MagDiff (signal → detectability) [Figure 4]
  - logFx ~ MagDiff + logd (signal + noise → observed flux) [Figure 5]

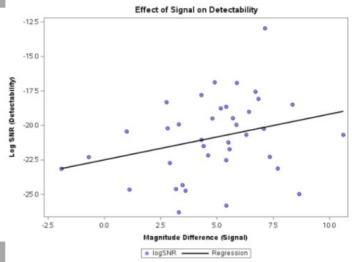



Figure 1

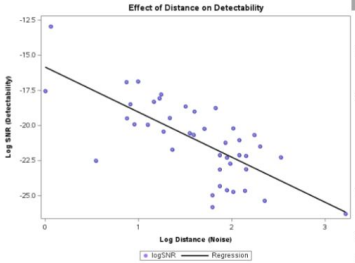



Figure 2

Pearson Correlation Coefficients
Prob > |r| under H0: Rho=0
Number of Observations

| | MagDiff | logd | logSNR |
|---|---|---|---|
| MagDiff | 1.00000<br>48 | -0.19206<br>0.2480<br>38 | 0.28920<br>0.0782<br>38 |
| logd | -0.19206<br>0.2480<br>38 | 1.00000<br>43 | -0.72932<br><.0001<br>42 |
| logSNR | 0.28920<br>0.0782<br>38 | -0.72932<br><.0001<br>42 | 1.00000<br>42 |

Figure 3

Analysis of Variance

| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
|---|---|---|---|---|---|
| Model | 1 | 25.36367 | 25.36367 | 3.29 | 0.0782 |
| Error | 36 | 277.88900 | 7.71914 | | |
| Corrected Total | 37 | 303.25267 | | | |

| | | | |
|---|---|---|---|
| Root MSE | 2.77833 | R-Square | 0.0836 |
| Dependent Mean | -20.87761 | Adj R-Sq | 0.0582 |
| Coeff Var | -13.30772 | | |

Parameter Estimates

| Variable | DF | Parameter Estimate | Standard Error | t Value | Pr > \|t\| |
|---|---|---|---|---|---|
| Intercept | 1 | -22.49068 | 0.99751 | -22.55 | <.0001 |
| MagDiff | 1 | 0.33132 | 0.18278 | 1.81 | 0.0782 |

Figure 4

Analysis of Variance

| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
|---|---|---|---|---|---|
| Model | 2 | 27.27645 | 13.63823 | 3.39 | 0.0450 |
| Error | 35 | 140.69843 | 4.01996 | | |
| Corrected Total | 37 | 167.97488 | | | |

| | | | |
|---|---|---|---|
| Root MSE | 2.00498 | R-Square | 0.1624 |
| Dependent Mean | -17.71485 | Adj R-Sq | 0.1145 |
| Coeff Var | -11.31809 | | |

Parameter Estimates

| Variable | DF | Parameter Estimate | Standard Error | t Value | Pr > \|t\| |
|---|---|---|---|---|---|
| Intercept | 1 | -17.03042 | 1.17975 | -14.44 | <.0001 |
| MagDiff | 1 | 0.18051 | 0.13440 | 1.34 | 0.1879 |
| logd | 1 | -0.98857 | 0.51158 | -1.93 | 0.0614 |

Figure 5

## RESULTS

**Correlation results:**

- MagDiff ↔ logSNR: r = 0.29, p = 0.078 (positive, not significant)
- logd ↔ logSNR: r = −0.73, p < 0.0001 (negative, strong)

**Regression: logSNR ~ MagDiff**

- F = 3.29, p = 0.078 → trend in expected direction, but not strong evidence
- $R^2$ = 0.084 → explains 8% of variance
- β = 0.331

**Regression: logFx ~ MagDiff + logd**

- F = 3.39, p = 0.045 → significant
- $R^2$ = 0.162 → explains 16% of variance
- MagDiff: β = 0.18, p = 0.188 → positive, not significant
- logd: β = −0.99, p = 0.061 → negative, marginally significant

## CONCLUSIONS

Larger outbursts (MagDiff) tend to increase X-ray detectability & flux, but the effects were modest. Distance (logd) is the dominant factor affecting observed flux, consistent with SDT framing (noise reduces detectability). My analysis suggests that signal contributes to X-ray emission but is partially masked by observational noise (distance). So while my findings are consistent with SDT, I am still limited by sample size & missing data.

## FUTURE DIRECTIONS

- Increase sample size with additional black hole catalogs to improve results
- Include other potential "noise" factors, or find a more fitting variable for that
- Go further into SDT framework and try to generalize this to other areas and topics
  - Can we learn anything from space detection to utilize that for other places that can rely on SDT?

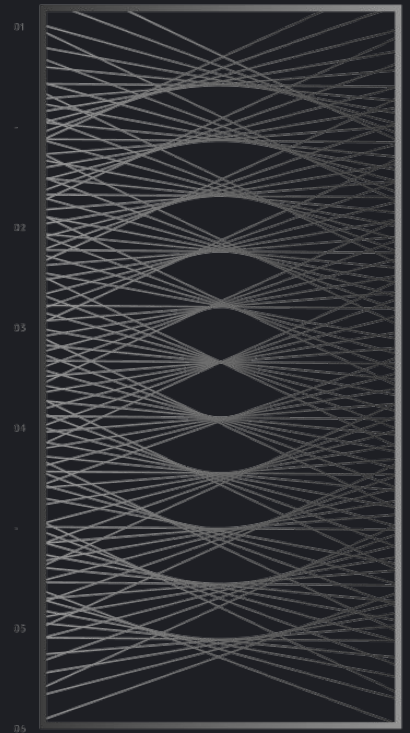



# SIGNAL DETECTION THEORY IN BLACK HOLES

AJ SAVARESE

Extracting Information From the Edge of Spacetime

INTRO



# WHAT IS SDT?

Signal Detection Theory is about how we distinguish real signals from noise, and how noise can make detection difficult even when the signal is strong.

# HYPOTHESIS OF STUDY

- Black holes can produce bursts of energy that we can detect
  - These are “outbursts” and “x-ray flux”
- See if brighter outbursts are associated with stronger flux
  - This is our signal
- Also need to account for the distance
  - This is our noise

**Stronger outbursts = Stronger detectability**
**Greater distance = Lower detectability**

< METHODS >

# DATA PREPROCESSING

## DATASET

BlackCAT black hole catalogue

## OBSERVATIONS

Only 73 in this set

## CREATINGS VARS.

Represent signal & noise

## HOW TO MEASURE

Making this interpretable



# VARIABLES OF IMPORTANCE

< METHODS >

| VARIABLE | CONCEPT | UNITS |
|---|---|---|
| MAGNITUDE DIFFERENCE (MAGDIFF) | Difference between brightness during an outburst and brightness at rest (MagOut - MagQui) | Magnitude (AB) |
| LOG X-RAY FLUX & LOG DISTANCE (LOGFX & LOGD) | Log-transformed flux & distance to make the data more balanced and easier to analyze | Flux = erg per second per square centimeter<br>D = Kiloparsecs (kpc) |
| LOG SIGNAL-TO-NOISE RATIO (LOGSNR) | Combined signal and noise into one measure of detectability (logFx - 2*logd) | N/A (My relative measure) |

< ANALYSIS >

| Pearson Correlation Coefficients<br>Prob > \|r\| under H0: Rho=0<br>Number of Observations | | | |
|---|---|---|---|
| | MagDiff | logd | logSNR |
| MagDiff | 1.00000<br><br>48 | -0.19206<br>0.2480<br>38 | 0.28920<br>0.0782<br>38 |
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| logSNR | 0.28920<br>0.0782<br>38 | -0.72932<br><.0001<br>42 | 1.00000<br><br>42 |

# PEARSON R CORRELATION

MagDiff ↔ logSNR
r = 0.29, p = 0.078
(positive, not significant)

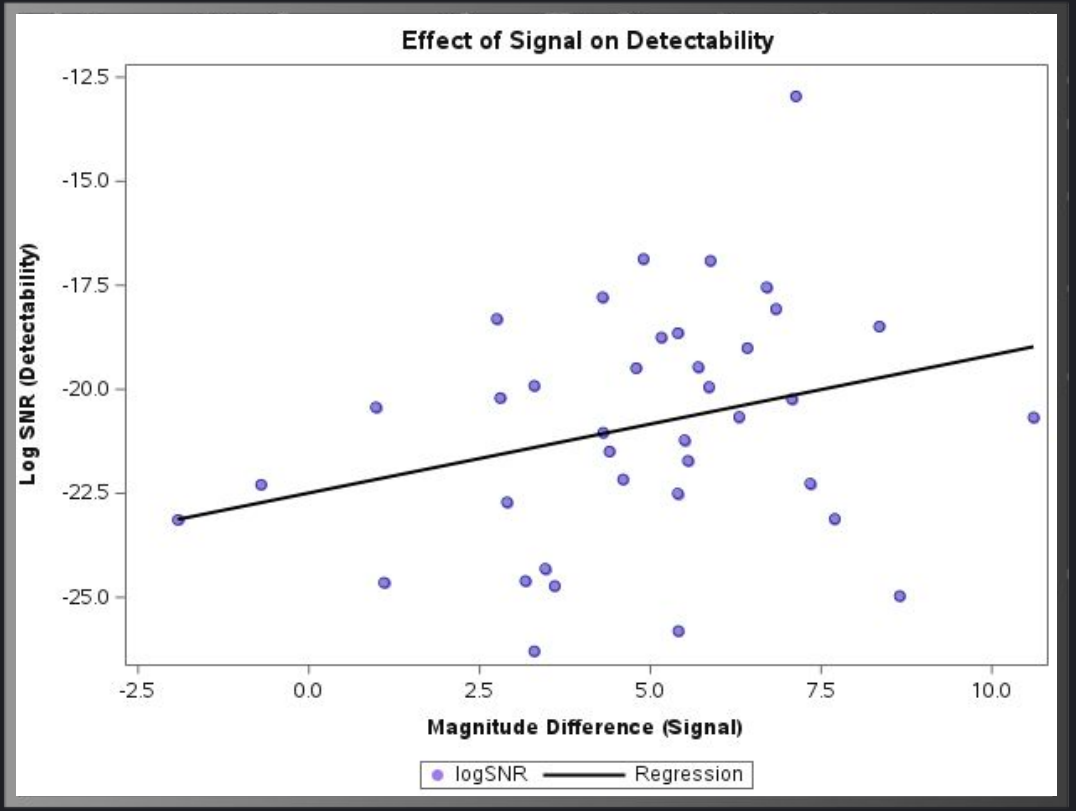
logd ↔ logSNR
r = −0.73, p < 0.0001
(negative, strong)

< ANALYSIS >

# MAGDIFF VS LOGSNR

r = 0.29

p = 0.078

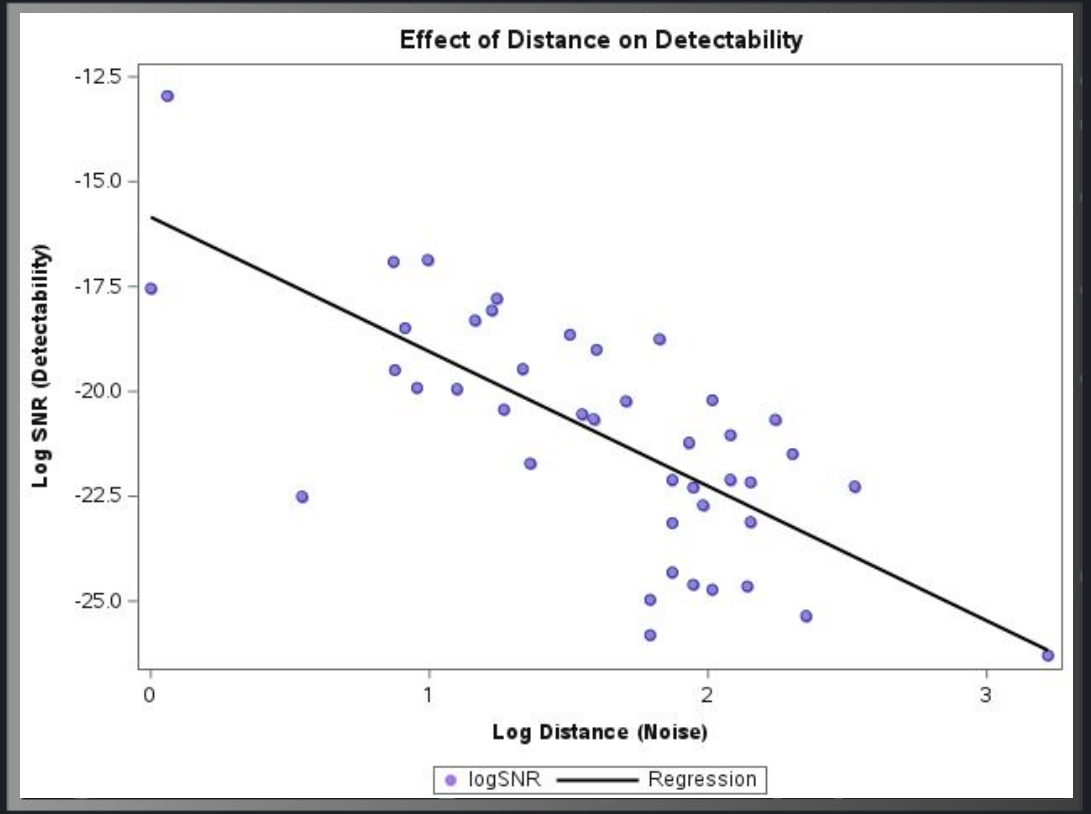


< ANALYSIS >

# LOGD VS LOGSNR

r = -0.73

p < 0.0001



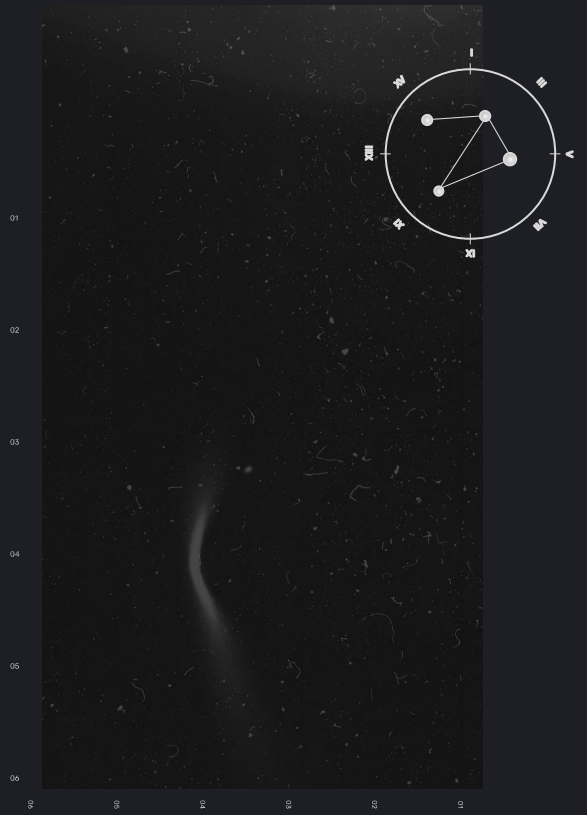
# REGRESSION RESULTS

< RESULTS >

| LOGSNR ~ MAGDIFF |
|---|
| F = 3.29 |
| p = 0.078 |
| $R^2$ = 0.084 |
| β = 0.331 |

| LOGFX ~ MAGDIFF + LOGD | |
|---|---|
| F = 3.39 | |
| p = 0.045 | |
| $R^2$ = 0.162 | |
| **MagDiff** | **logd** |
| β = 0.18 | β = -0.99 |
| p = 0.188 | p = 0.061 |

CONCLUSION



# WHAT CAN BE CONCLUDED?

- Larger outbursts (signal) associated with slightly higher X-ray detectability & flux
  - Effect of signal isn’t dominant factor
- Distance (noise) is strongest predictor of observed flux
- Results are consistent with Signal Detection Theory

# FUTURE DIRECTIONS

< FUTURE >



## THE DATA

More catalogues could help + find other potential “noise” factors



## THE WHOLE

Go further into SDT framework + generalizing for other places that can rely on SDT



## THE KNOWLEDGE

Those more familiar with astronomy may fair better + overall better understanding